

DSAA1001

Module 1 Overview & Logistics

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HKUST(GZ)

2 September 2026

Outline

Course Logistics

Module 1: New Learning Method

Using the Learning Sheet

Using AI Tools

Labs

Summary

Schedule & Location

Lecture

- Mon/Wed, 10:30–11:50
- 2–30 September 2026
- E1-134
- Lecture 9 ends at 11:00

Lab

- Every Monday, 12:00–12:50
- E1-134

Notes

- Module 1 uses active, self-paced learning supported by the course AI platform and the teaching team.
- Bring a laptop or tablet to Module 1 lectures and labs.

Three Course Modules

Maths & Algorithms

Prof. Wei Wang

Data

Prof. Lei Li

Machine Learning

Prof. Yongqi Zhang

Module 1: 2–30 September 2026 · **Assessment:** 20% of the course

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Why This Course Is Worth the Effort

According to **Expectancy-Value Theory (EVT)**, effort rises when value is high, success feels attainable, and cost is acceptable.

Utility value

- Required for DSA/AI paths.
- Useful across math, algorithms, data, systems, and ML.

Intrinsic value

- Keep curiosity alive.
- Independent learning is a durable AI-era advantage.

Attainment value

- We provide sheets, tests, support, and AI tools.
- You convert them into competence and marks.

Cost

- Almost no extra cost: you must study anyway.
- In-class work can beat struggling alone; DSAA3071 feedback supports this.

Why This New Approach?

Pacing

- One lecture speed cannot fit everyone.
- Some are lost; others wait.

AI-era self-learning

- Practice attention, retrieval, and repair.
- Productive struggle is part of learning.

Reading beats passive listening

- Reading exposes gaps.
- Listening can create false mastery.

Use required time well

- You must learn anyway.
- Structured class time should make it faster.

How It Works: Session Format

Test

- Lectures 3, 5, 7, and 9.
- Closed-book test in the first 30 minutes.

Learning time

- Self-paced learning-sheet work.
- Active reading, problem solving, and repair.
- Course AI platform + instructor support.

Test coverage

- Each test covers the preceding two lectures.
- Example: the Lecture 5 test covers Lectures 3 and 4.
- Validation sheets support preparation.

Additional activities

- Discussion.
- Mini-lectures or mini-research projects.

Assessment Structure

Module 1 assessment: 20%

- Four in-class tests: **16%**
- Mandatory learning-sheet answers: **4%**
- The four-test average determines the 16% test component.

Test conditions

- Closed-book.
- Instructor-provided cheat sheet allowed.
- Ordinary calculator allowed.
- No AI tools.

Validation sheets use similar question types and formats, but test content and depth may vary.

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Six Key Strategies

1. Focus on Tiers 1 + 2

- Tier 3 is optional and not tested.
- Do not try to finish everything.

2. Follow the logic

- Techniques are threaded together.
- Do not memorize isolated tricks.

3. Use the devices

- Examples, diagrams, and exercises teach.
- Do not treat them as decoration.

Six Key Strategies (continued)

4. Ask, and be ready to be asked

- I will circulate.
- Be ready to explain your thinking.

5. Ask for help when stuck

- Productive struggle is good.
- Spinning wheels is not.

6. Answer questions as you learn

- After reasonable thinking, write down your attempt.
- Marks are not based on whether your answers are correct.

Use all resources: materials, instructor, peers, and AI tools.

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AI Tool Requirements

Requirements

- Use the course-specific AI platform for learning activities.
- Submit mandatory learning-sheet answers through the platform.
- Use the course platform rather than external AI tools in class.
- Do not use AI tools during tests.

Purpose

- Clarify concepts and request hints.
- Generate examples and debug code.
- Judge **when** and **how** to use AI.
- Keep responsibility for your reasoning.

Verify rather than trust: the learning sheet may contain AI-generated errors, and AI-generated answers may also be wrong.

How to Learn with AI Without Outsourcing Thinking

Read first

Interpret definitions, examples, and problem statements before asking AI.

Repair and transfer

Rewrite the solution yourself; test it on a nearby variant.

Optional resources

- Feynman Technique: explain simply; find gaps; repair; repeat.
- Socratic questioning: clarify assumptions, reasons, evidence, and alternatives.
- Intuition & mental models: build concrete pictures before formal symbols.
- How To Become Dangerously Self-Educated (with AI): <https://www.youtube.com/watch?v=VeU6gScy92s>

Ask for diagnosis

Prefer “where is my reasoning wrong?” to “give me the answer.”

Own the attainment

AI accelerates feedback; it cannot replace retrieval, judgment, or curiosity.

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Lab Approach

Schedule: every Monday, 12:00–12:50, in E1-134.

Goal: solve the lab problem during lab time.

Work mode

- Individual or collaborative, as appropriate.
- Ask questions and debug with support.

Purpose

- Apply learning-sheet concepts.
- Turn theory into experiments and working code.

Labs are where theory meets practice.

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Key Takeaways

Logistics

- Lectures: Mon/Wed, 10:30–11:50
- Labs: Monday, 12:00–12:50
- E1-134
- Bring a laptop or tablet

Learning and assessment

- Self-paced sheet work
- Four tests: 16%; mandatory answers: 4%
- Tiers 1 + 2 are the test priorities
- Course AI platform for learning; no AI in tests

Use the new method to learn how to learn effectively.

Questions?

“一切教育本质上都是自我教育，一切学习本质上都是自学”

“Education is ultimately self-education, and learning is ultimately self-learning.”

— Guoping Zhou

Source: <https://www.sinoss.net/c/2016-03-09/524538.shtml>

Ready to start learning!